

ENERGY: GRID PEAK-LOAD PREDICTION

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ABSTRACT - The increasing integration of renewable energy sources into modern power systems introduces challenges due to their intermittent and variable nature, particularly during periods of high electricity demand. This study presents a data-driven framework for analyzing historical electricity demand and renewable generation patterns to identify potential renewable-demand mismatch periods and reliability windows. Electricity-system data obtained from the ENTSO-E Transparency Platform is analyzed at 15-minute intervals over approximately three years, incorporating electricity load, solar generation, onshore and offshore wind generation, and hydro generation. The methodology involves data preprocessing, temporal feature extraction, renewable-generation aggregation, renewable coverage estimation, renewable-demand gap analysis, demand-spike identification, and peak-load prediction using machine-learning techniques. The analysis focuses on determining whether periods of peak electricity demand coincide with reduced renewable-energy availability and identifying recurring time windows associated with increased grid stress. The proposed framework provides a basis for assessing the relationship between electricity demand and renewable generation rather than analyzing these factors independently. The resulting insights can support grid preparedness, demand-side management, flexible generation planning, and future energy-storage optimization. The study contributes to the development of resilient and renewable-integrated energy systems and supports Sustainable Development Goal (SDG) 7: Affordable and Clean Energy, SDG 9: Industry, Innovation and Infrastructure, SDG 12: Responsible Consumption and Production, and SDG 13: Climate Action.

I. INTRODUCTION

A. Background and Importance

The increasing adoption of renewable energy sources such as solar and wind is transforming modern electricity systems. However, renewable generation is inherently variable because it depends on environmental conditions such as solar irradiance and wind availability. At the same time, electricity demand changes throughout the day and across seasons, with certain periods experiencing significant demand spikes. This variability creates a challenge for maintaining a continuous balance between electricity generation and demand.

The combination of **high electricity demand and low renewable generation** can create a renewable-demand mismatch and increase pressure on grid reliability. Conversely, periods of high renewable generation and relatively low demand may result in excess energy that could potentially be utilized through energy-storage systems. Therefore, jointly analyzing electricity demand and renewable generation is important for

identifying periods when the grid may experience increased stress. This study focuses on identifying such recurring periods,

referred to as **renewable reliability windows**, and using these insights to support peak-load prediction and grid-risk assessment.

B. Limitations of Existing Work

Existing studies on electricity-load forecasting generally focus on predicting future demand using historical load patterns, while renewable-energy studies often examine the variability and forecasting of individual sources such as solar and wind. Although these approaches provide useful insights, they may not fully address the relationship between **demand peaks and renewable-energy availability**. In particular, analyzing load and renewable generation separately can overlook periods when high electricity demand coincides with low renewable generation, which may create increased pressure on grid reliability.

Another limitation is that identifying such demand-renewable mismatches requires **high-resolution temporal analysis**. Aggregating data into longer time intervals can mask short-duration demand spikes or rapid changes in renewable generation. Furthermore, conventional peak-load prediction does not necessarily indicate whether sufficient renewable generation is available during the predicted peak period. Therefore, there is a need for an integrated approach that considers **electricity demand, renewable generation, and their temporal relationship** to identify potential reliability windows and high-risk periods. This study addresses this limitation by jointly analyzing historical load and renewable-generation data and using the resulting patterns as a foundation for peak-load prediction and grid-risk assessment.

C. Contributions, Significance and Implications

This study contributes an integrated framework for analyzing electricity demand and renewable-energy generation together to identify potential **renewable-demand mismatch periods and reliability windows**. Unlike an approach that considers peak-load prediction alone, the proposed framework examines whether renewable generation is sufficient relative to demand during periods of increased electricity consumption. Using high-resolution ENTSO-E data, the study derives indicators such as total renewable generation, renewable coverage, and the renewable-demand gap to characterize grid conditions across different time periods.

The significance of this work lies in providing a data-driven basis for understanding the relationship between **demand spikes and renewable availability**. The identified patterns can support the prediction of high-demand periods and the assessment of potential grid-risk conditions. From a practical perspective,

these insights may assist in planning flexible generation, demand-side management, and energy-storage deployment during periods of potential system stress. The framework also provides a foundation for future work on **storage optimization using reinforcement learning**, where surplus renewable energy could be stored during favorable periods and utilized during high-demand, low-renewable windows. Thus, the proposed approach can contribute to more reliable and efficient integration of renewable energy into modern power systems.

II. LITERATURE REVIEW

Electricity load forecasting is an important component of power-system planning and operation because accurate estimation of future demand supports generation scheduling, system expansion, demand management, and reliable operation of the electricity grid. The forecasting problem is challenging because electricity demand exhibits nonlinear, non-stationary and seasonal characteristics and is influenced by temporal, environmental and socioeconomic factors. Consequently, research in electricity-load forecasting has progressed from conventional statistical time-series techniques toward machine-learning and hybrid forecasting frameworks capable of representing nonlinear relationships. The literature relevant to peak-load forecasting demonstrates a progression from long-term forecasting models to short-term machine-learning approaches and more sophisticated multivariate hybrid architectures.

A. Hybrid Approaches for Long-Term Peak-Load Forecasting

Kazemzadeh, Amjadian and Amraee [1] proposed a hybrid data-mining-driven methodology for long-term electricity peak-load and energy-demand forecasting. The study addressed the forecasting of yearly peak load and total energy demand for the Iranian national electricity system. The authors developed a Support Vector Regression (SVR) model and employed Particle Swarm Optimization (PSO) to optimize the parameters of the SVR model and the dimension of the input samples. The proposed PSO-SVR approach was subsequently combined with conventional time-series forecasting through Auto-Regressive Integrated Moving Average (ARIMA) and Artificial Neural Network (ANN) models. The hybrid framework prioritized the forecasting approaches according to their observed errors and was evaluated using real-world data from the Iranian national grid. The authors reported that the proposed PSO-SVR and hybrid approaches provided more accurate forecasts than the individual ARIMA and ANN approaches for the considered long-term forecasting problem.

The significance of this study lies in demonstrating that combining statistical and machine-learning approaches can improve the representation of complex load behaviour. Conventional ARIMA-based methods are effective for temporal dependencies but may have difficulty representing nonlinear relationships, whereas machine-learning approaches can capture more complex patterns. The hybrid strategy therefore provides a useful foundation for subsequent research in electricity-load forecasting. However, the study is primarily concerned with long-term yearly peak-load and total energy-demand forecasting, with a forecasting horizon extending several years into the future. Consequently, its methodology is not directly designed for high-resolution short-term identification of

individual peak-demand intervals or for analysing the relationship between peak demand and variable renewable-generation availability.

B. Machine-Learning-Based Short-Term Peak-Load Forecasting

Kumar and Pal [2] investigated machine-learning-based electricity-load forecasting for peak-demand control in a smart-grid environment. The study focused on the Jajpur electricity grid in Odisha, India, and used historical electricity-load data together with weather-related variables. The authors evaluated a broad range of regression and machine-learning techniques, including linear regression, decision-tree-based methods, Support Vector Machines (SVM), Gaussian Process Regression (GPR), and neural-network models. The study used hourly load observations, with twelve months of data used for training and subsequent months used for validation and testing. Among the evaluated models, the Rational Quadratic Gaussian Process Regression model achieved the lowest reported RMSE of 2.0466, indicating strong predictive performance for the selected dataset and evaluation setting.

An important aspect of this work is its emphasis on short-term forecasting for peak-demand control rather than exclusively long-term capacity planning. The study also examined relationships between electrical demand and environmental variables such as temperature, humidity, wind, cloud and precipitation-related factors, demonstrating the importance of incorporating explanatory variables into electricity-load forecasting. Nevertheless, the study identifies several practical limitations. The authors note that the available training data were limited and that a small dataset may lead to underfitting or overfitting. They also report a trade-off between forecasting accuracy and computational complexity: although Rational Quadratic GPR achieved the lowest RMSE, it required greater computational time than simpler approaches such as SVM. Furthermore, the model is primarily designed to predict electrical load and does not explicitly integrate renewable-generation adequacy into the peak-demand assessment.

C. Multivariate Hybrid Models for Day-Ahead Peak-Load Forecasting

Huang et al. [3] proposed a Multivariate Empirical Mode Decomposition–Particle Swarm Optimization–Support Vector Regression (MEMD-PSO-SVR) framework for day-ahead peak-load forecasting. The study addressed the nonlinear and non-stationary characteristics of electricity-load data by incorporating multivariate time-frequency decomposition before applying the forecasting model. MEMD was used to decompose historical peak-load data and influential variables into intrinsic mode functions while retaining information across multiple variables and time-frequency components. PSO was then employed to optimize the parameters of the SVR model. The proposed methodology was evaluated using two real-world electricity-load datasets corresponding to the New South Wales and Victoria electricity systems in Australia.

The study demonstrates the potential of combining advanced preprocessing, optimization and machine-learning techniques for improving day-ahead peak-load forecasting. In particular, the use of MEMD addresses the time-frequency characteristics of nonlinear and non-stationary load data, while PSO provides automated parameter optimization for SVR. The authors report that the proposed hybrid methodology provides a promising alternative for day-ahead peak-load forecasting and emphasize its potential application to power-system planning and operation. However, the resulting methodology involves multiple processing and optimization stages, increasing methodological complexity compared with a simpler feature-engineered machine-learning framework. More importantly, the primary objective remains accurate day-ahead peak-load prediction. Although multiple load-related and influential variables are incorporated, the framework does not directly transform peak-load forecasts into an assessment of whether renewable generation is adequate during those peak periods.

D. Comparative Analysis of Existing Approaches

The reviewed studies demonstrate a clear evolution in electricity peak-load forecasting. Kazemzadeh et al. [1] demonstrated the effectiveness of combining statistical and machine-learning techniques for long-term yearly peak-load and energy-demand forecasting. Kumar and Pal [2] shifted the focus toward short-term peak-demand prediction using a broad set of machine-learning regression techniques and environmental variables. Huang et al. [3] further advanced the forecasting methodology through multivariate time-frequency decomposition and optimization-based hybrid modelling for day-ahead peak-load prediction. These studies collectively demonstrate that machine-learning and hybrid methodologies can address the nonlinear, non-stationary and complex characteristics of electricity demand more effectively than relying exclusively on conventional statistical approaches.

However, the reviewed studies predominantly formulate the forecasting objective around the prediction of electricity demand or peak-load magnitude. Consequently, model performance is primarily interpreted through forecasting accuracy rather than through the operational context in which a predicted peak occurs. This distinction becomes increasingly relevant in electricity systems with substantial renewable-energy penetration. A high-demand period may represent different levels of operational concern depending on the amount of renewable generation available at the same time. For example, a demand peak accompanied by strong wind and solar generation may impose a different balancing requirement from a similar demand peak occurring when renewable generation is relatively low. The reviewed forecasting frameworks do not explicitly establish such a demand–renewable adequacy relationship within the peak-load forecasting process. This observation motivates the integrated approach adopted in the present study.

E. Summary of Literature

Overall, the literature establishes that electricity peak-load forecasting has benefited from the progression from conventional statistical techniques toward machine-learning and hybrid approaches. Existing studies have explored different forecasting horizons, datasets and methodological strategies, including ARIMA–ANN–PSO–SVR hybrid modelling, Gaussian Process Regression, Support Vector Machines, neural networks

and MEMD–PSO–SVR. The studies also demonstrate the importance of incorporating historical load information and relevant explanatory variables to improve predictive performance.

Despite these advances, the reviewed literature primarily treats load forecasting and renewable-generation adequacy as separate analytical concerns. The literature therefore provides a strong foundation for predicting electricity demand but leaves an opportunity to extend the analysis from simply estimating *when and how much electricity will be demanded* toward determining *whether renewable generation is sufficiently available during potentially critical demand periods*. The present study addresses this opportunity by combining high-resolution electricity-load prediction, peak-demand identification and renewable-generation adequacy analysis within a unified framework for identifying potentially critical grid-reliability windows.

III. RESEARCH GAP

The reviewed literature demonstrates significant progress in electricity peak-load forecasting through statistical, machine-learning, and hybrid approaches. Existing studies have successfully addressed long-term peak-load forecasting, short-term demand prediction, and day-ahead peak-load forecasting using methods such as ARIMA, Support Vector Regression, Gaussian Process Regression, neural networks, and multivariate hybrid models. However, the primary objective of these approaches remains the accurate prediction of electricity demand or peak-load magnitude.

A key limitation identified from the reviewed studies is that **peak-load forecasting and renewable-generation adequacy are generally treated as separate analytical problems**. A predicted high-demand period does not necessarily represent the same level of grid stress under different renewable-generation conditions. For example, a demand peak occurring during strong wind and solar generation may present a different operational condition from an equivalent demand peak occurring when renewable generation is low. The reviewed studies do not explicitly incorporate this demand–renewable relationship into the identification and assessment of critical peak periods.

Furthermore, existing studies reviewed in this work do not specifically focus on identifying **recurring high-demand and low-renewable time windows** as an indicator of potential grid reliability risk. Consequently, there is an opportunity to extend conventional peak-load forecasting from simply answering “**when and how much electricity will be demanded?**” to also address “**how much renewable generation is available during that critical demand period?**” This study addresses this gap by integrating high-resolution load analysis, renewable-generation assessment, peak-demand identification, and renewable-demand mismatch analysis within a unified framework for identifying potential **renewable reliability windows** and supporting grid-risk assessment.

IV. PROBLEM STATEMENT

The transition toward renewable-energy-based electricity generation is changing the operating characteristics of modern power systems. Unlike conventional generation, renewable sources such as solar and wind are strongly dependent on

naturally varying conditions, while electricity consumption follows its own temporal patterns and can increase sharply during peak-demand periods. This creates a fundamental challenge for electricity-system planning: the occurrence of a demand peak does not necessarily represent the same level of operational concern at every point in time, because the availability of renewable generation can differ substantially between otherwise similar demand conditions.

Existing electricity-load forecasting research has made significant progress in predicting future demand and identifying peak-load behaviour through statistical, machine-learning and hybrid approaches. However, forecasting the magnitude of electricity demand alone provides limited information about the relationship between a demand peak and the generation resources available to satisfy it. Similarly, examining renewable generation independently does not reveal whether periods of low renewable availability coincide with periods of elevated demand. As a result, there is a need to move beyond treating electricity-demand forecasting and renewable-generation analysis as separate problems and instead examine their combined temporal behaviour.

The specific problem addressed in this study is therefore the identification and characterization of periods in which elevated electricity demand coincides with comparatively low renewable-generation contribution. Such conditions are important because they may indicate greater dependence on non-renewable generation, imports, flexible generation or future energy-storage resources. The study seeks to establish a data-driven approach for distinguishing ordinary demand fluctuations from demand conditions in which renewable availability may be relatively insufficient. In this context, the term reliability window refers to a potentially critical period arising from the interaction between electricity-demand intensity and renewable-generation availability, rather than a complete physical measurement of power-grid reliability.

The study consequently focuses on answering a central question:

How can electricity-demand behaviour and renewable-generation availability be jointly analyzed to identify potentially critical periods where high demand coincides with comparatively low renewable contribution?

Addressing this problem can provide a more meaningful interpretation of peak-demand behaviour than load forecasting alone and can establish an analytical foundation for future applications involving demand response, flexible generation and energy-storage optimization.

V. OBJECTIVES

The primary objective of this study is to develop a data-driven framework for analyzing electricity demand and renewable-generation patterns to identify potential renewable-demand mismatch periods and reliability windows. The specific objectives are:

1. **To analyze historical electricity demand patterns** and identify peak-load periods, including their temporal and seasonal characteristics.
2. **To analyze renewable-generation variability** across solar, onshore wind, offshore wind, and hydro generation using high-resolution ENTSO-E data.
3. **To quantify the relationship between electricity demand and renewable generation** using indicators such as total renewable generation, renewable coverage, and the renewable-demand gap.
4. **To identify recurring reliability windows** in which high electricity demand coincides with relatively low renewable-generation availability.
5. **To develop and evaluate peak-load prediction models** using historical load and relevant temporal and renewable-generation features.
6. **To assess potential grid-risk conditions** by combining demand peaks with renewable-generation availability during corresponding periods.
7. **To establish a foundation for future energy-storage optimization**, where surplus renewable generation can potentially be stored and utilized during high-demand, low-renewable periods.

VI. RESEARCH METHODOLOGY

The proposed methodology integrates electricity-load analysis, renewable-generation assessment, peak-demand identification, machine-learning-based load prediction, demand–renewable mismatch analysis, and reliability-window identification into a unified data-driven framework. The methodology is designed to investigate not only how electricity demand varies over time, but also whether periods of elevated demand coincide with comparatively inadequate renewable-generation availability. The overall workflow follows an Input–Process–Output architecture, in which high-resolution German electricity-system observations are validated, transformed into analytical features, analyzed through statistical and machine-learning techniques, and finally converted into load forecasts, demand–renewable indicators and potentially critical reliability windows.

A. Dataset Acquisition and Data Validation

The study uses a multivariate time-series dataset derived from the ENTSO-E Transparency Platform. The dataset contains 105,216 observations and 22 variables, recorded at a 15-minute temporal resolution over approximately three years, from 31 December 2022 23:00 UTC to 31 December 2025 22:45 UTC. The dataset contains actual electricity load, load forecasts, solar generation, onshore and offshore wind generation, hydro generation and temporal characteristics. It also contains derived indicators including total renewable generation, renewable coverage, renewable-generation gap, peak-demand flag, forecast error and risk-related variables.

Before analysis, the dataset undergoes a validation and preprocessing stage. The timestamp variable is converted into a standardized datetime format and all observations are arranged chronologically. The dataset is checked for missing values, duplicate records, timestamp continuity, invalid numerical values and consistency of derived variables. Since the observations are expected to occur at 15-minute intervals, consecutive timestamps are examined to ensure temporal continuity. The renewable-generation total is also independently reconstructed from its component sources and compared with

the supplied renewable-total variable. This validation ensures that subsequent lag-based, rolling and time-series analyses operate on a structurally consistent dataset.

B. System Workflow

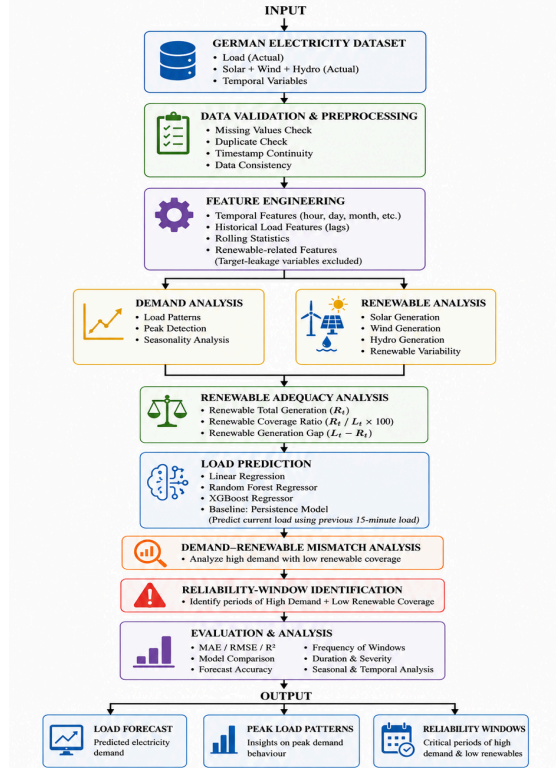


Image 1: Flowchart of our system workflow

C. Data Preprocessing and Feature Engineering

Temporal and historical characteristics are extracted to represent recurring electricity-demand behaviour. Existing variables such as hour, day of week, day of year, month and weekend indicator are used to capture daily, weekly and seasonal patterns. Additional historical load features are constructed using lagged observations, including L_{t-1} , L_{t-2} , L_{t-4} , L_{t-96} and L_{t-672} , corresponding approximately to the previous 15 minutes, 30 minutes, one hour, one day and one week. Rolling statistics such as moving averages and rolling standard deviations may also be calculated using only previously observed values. A general moving-average feature is represented as

$$MA_k(t) = \frac{1}{k} \sum_{i=1}^k L_{t-i}$$

This historical formulation prevents future observations from being incorporated into prediction features.

A specific target-leakage prevention procedure is applied during feature selection. Variables that are directly calculated using the target electricity load at the same timestamp, such as renewable-generation gap, renewable coverage and load-forecast error, are not automatically supplied to the load-prediction models. For example,

$$G_t = L_t - R_t$$

contains the actual target L_t . Using such a variable directly to predict L_t would provide information derived from the answer itself and produce an unrealistically optimistic model. Therefore, predictive features are restricted to variables that would be available independently at the defined prediction time.

D. Renewable-Generation Analysis

Renewable generation is analyzed by combining the renewable sources available in the dataset. Total renewable generation is represented as

$$R_t = S_t + Won_t + Woff_t + Hrun_t + Hres_t,$$

where S_t represents solar generation, Won_t and $Woff_t$ represent onshore and offshore wind generation, and $Hrun_t$ and $Hres_t$ represent run-of-river and reservoir hydro generation, respectively. The calculated value is compared with the dataset-provided renewable-total variable to verify consistency. Renewable generation is subsequently examined across different hours, weekdays, months, seasons and demand conditions to identify temporal patterns and variations in renewable availability.

E. Renewable-Adequacy and Demand-Renewable Mismatch Analysis

Renewable adequacy is evaluated relative to simultaneous electricity demand rather than by considering renewable generation independently. Two principal mathematical indicators are used. The Renewable Coverage Ratio (RC) is defined as

$$RC_t = \frac{R_t}{L_t} \times 100,$$

where R_t is total renewable generation and L_t is electricity load at time t . The ratio indicates the percentage of simultaneous demand represented by renewable generation. The second indicator is the Renewable Generation Gap, defined as

$$G_t = L_t - R_t,$$

A positive G_t indicates that electricity demand exceeds renewable generation, whereas a negative value indicates that renewable generation exceeds the observed demand. These indicators are compared during peak and non-peak periods to determine whether high-demand conditions systematically exhibit different renewable-adequacy characteristics.

F. Peak-Demand Identification and Demand-Renewable Matrix

Peak demand is identified using the upper region of the observed load distribution. An independent threshold is calculated using the 90th percentile:

$$P_t = \begin{cases} 1, & L \geq Q_{90}(L) \\ 0, & L < Q_{90}(L). \end{cases}$$

The independently calculated peak criterion is compared with the dataset-provided `peak_flag` rather than treating the supplied flag as unquestionable ground truth. Demand and renewable availability are subsequently categorized into four conditions: low demand–high renewable availability, low demand–low renewable availability, high demand–high renewable availability, and high demand–low renewable availability. The final category represents the principal condition investigated by this study because it combines elevated electricity demand with comparatively limited renewable contribution.

G. Reliability-Window Identification

A potentially critical reliability window is defined as a period in which high electricity demand coincides with comparatively low renewable coverage. The baseline data-driven criterion is expressed as

$$C_t = (L_t \geq Q_{90}(L)) \wedge (RC_t < Q_{25}(RC)),$$

where $Q_{90}(L)$ is the 90th percentile of electricity load and $Q_{25}(RC)$ is the lower quartile of renewable coverage. The identified observations are subsequently grouped into potentially continuous windows and analyzed according to frequency, duration, demand magnitude, renewable coverage, renewable-generation gap, hour of day, day of week, month and seasonal occurrence. The term *potentially critical* is deliberately used because the analysis identifies demand–renewable conditions rather than directly measuring complete physical grid reliability.

H. Machine-Learning-Based Load Prediction

The predictive component focuses on short-term electricity-load forecasting. Linear Regression, Random Forest Regression and XGBoost Regression are considered as candidate models representing linear, ensemble-tree and gradient-boosting approaches. The prediction target is actual electricity load L_t . Historical load features and temporal variables are used as predictive features, while the unverified forecast-related field is excluded from the predictive feature matrix. A persistence model was employed as the forecasting baseline, where the load at the immediately preceding 15-minute interval was used as the prediction for the current interval. This provides a simple but meaningful benchmark because short-term electricity demand exhibits strong temporal persistence. The purpose of this comparison is to determine whether the developed machine-learning models can provide competitive predictive performance while also supporting subsequent peak-demand and reliability-window analysis.

I. Experimental Design and Evaluation

Since the observations form a time series, the data are divided chronologically into training and testing periods rather than randomly shuffled. This ensures that future observations do not influence model training and provides a more realistic representation of forecasting performance. The candidate models are evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Coefficient of Determination (R²). MAE measures the average absolute prediction error,

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|,$$

while RMSE is defined as

$$RMSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2,$$

The coefficient of determination is calculated as

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2},$$

Lower MAE and RMSE and higher R² indicate better predictive performance. Model predictions are subsequently examined alongside renewable-generation conditions rather than evaluating forecasting accuracy in isolation.

J. Software, Computational Environment and Visualization

The analysis is implemented in Python using standard data-science and machine-learning libraries. Pandas is used for data loading, cleaning, transformation and time-series manipulation, while NumPy is used for numerical computation and mathematical feature construction. Matplotlib and Seaborn are used for exploratory data analysis and visualization. Scikit-learn provides the regression models, preprocessing utilities and evaluation metrics, while XGBoost is used for gradient-boosting-based regression. The workflow can be executed using a Python development environment or Jupyter Notebook. No specialized grid hardware is required because the study is based on historical data analysis and software-based experimentation; the computational hardware configuration used during execution can be documented separately in the experimental setup if required by the final submission format.

K. Overall Algorithm and Output

The complete algorithm proceeds sequentially from dataset acquisition → validation → preprocessing → temporal and historical feature engineering → renewable-generation aggregation → exploratory demand and renewable analysis → renewable-adequacy calculation → independent peak identification → demand–renewable classification → reliability-window identification → chronological ML training and testing → model evaluation → predicted-demand analysis. The resulting outputs include electricity-load forecasts, peak-demand characteristics, renewable-coverage and renewable-gap indicators, and potentially critical demand–renewable reliability windows. These outputs provide the analytical foundation for discussing grid-management implications and for the proposed future extension toward energy-storage optimization using reinforcement learning.

VII. DATASET

The study utilizes a multivariate time-series dataset representing the German electricity system, derived from the ENTSO-E Transparency Platform. The dataset contains 105,216 observations across 22 variables, recorded at 15-minute intervals

over approximately three years, from December 2022 to December 2025. This high temporal resolution enables the analysis of short-term variations in electricity demand and renewable generation that may not be captured through hourly or daily aggregation.

The dataset contains actual electricity load, load forecasts, solar generation, onshore and offshore wind generation, and hydroelectric generation. In addition to these primary measurements, the dataset includes temporal attributes such as hour, day of the week, day of the year, month and weekend indicators. The working dataset initially contained a forecast-related field; however, its provenance and forecasting alignment could not be independently verified. Therefore, this field was excluded from the final predictive feature matrix and was not used for model training, evaluation, or comparison. These variables enable the identification of daily, weekly and seasonal patterns in electricity consumption and renewable generation. The dataset also provides derived variables including total renewable generation, renewable coverage percentage, renewable-generation gap, peak-demand indicator and load-forecast error, which support the analysis of demand–renewable interactions.

The actual electricity load, represented by L_t , is the principal demand variable used for forecasting and peak-demand analysis. Renewable generation, represented by R_t , is obtained from the available solar, wind and hydro sources. The relationship between these two quantities is examined using renewable coverage and renewable-generation gap indicators. This allows the dataset to be analyzed not only in terms of electricity-demand behaviour, but also in terms of how renewable generation contributes to meeting demand at different points in time.

The dataset exhibits substantial variation in electricity demand and renewable generation across the observation period. Electricity load ranges from approximately 27.93 GW to 75.97 GW, with an average demand of approximately 52.83 GW. Preliminary comparison shows that peak-demand periods have an average load of approximately 68.50 GW, compared with 51.09 GW during non-peak periods. Peak periods also exhibit lower average renewable coverage (39.64%) compared with non-peak periods (47.82%) and a larger average renewable-generation gap (41.38 GW compared with 26.52 GW). These characteristics make the dataset suitable for investigating the relationship between demand peaks and renewable-generation availability.

Before analysis, the dataset is examined for missing values, duplicate observations, timestamp continuity and consistency of derived variables. The supplied dataset contains complete observations without missing values or duplicate records, providing a consistent basis for time-series analysis. Overall, the dataset provides the required combination of high-resolution electricity demand, renewable-generation measurements, and temporal information necessary to develop and evaluate the proposed demand–renewable reliability-window framework.

VIII. RESULTS

The proposed forecasting and renewable-reliability framework was evaluated using the three-year German electricity dataset

containing **105,216 observations at 15-minute intervals**. After data validation, feature engineering, and removal of unavailable lagged observations, the forecasting experiment employed a chronological split consisting of **83,635 training observations and 20,909 testing observations**. Three supervised learning models—Linear Regression, Random Forest, and XGBoost—were evaluated against a persistence baseline. The persistence model predicts the current load using the immediately preceding 15-minute load observation.

The models were evaluated using **Mean Absolute Error (MAE)**, **Root Mean Square Error (RMSE)**, and the coefficient of determination (R^2). Lower MAE and RMSE indicate smaller prediction errors, whereas a higher R^2 indicates that a larger proportion of the variation in electricity demand is explained by the model.

Model	MAE	RMSE	R^2
Persistence Baseline	506.90	670.42	0.994445
Linear Regression	289.44	398.98	0.998033
Random Forest	272.41	364.91	0.998354
XGBoost	254.36	341.28	0.998561

Table 1: Model Results

The **XGBoost model achieved the best overall forecasting performance**, obtaining an MAE of **254.36 MW**, RMSE of **341.28 MW**, and $R^2=0.998561$. Compared with the persistence baseline, XGBoost reduced MAE by approximately **49.8%** and RMSE by approximately **49.1%**. Random Forest was the second-best model, while Linear Regression provided a competitive linear benchmark.

The high R^2 values should be interpreted in the context of the **15-minute forecasting resolution** and the strong temporal persistence of electricity demand. Even the persistence baseline achieved $R^2=0.994445$, demonstrating that consecutive electricity-load observations are highly correlated.

A. Feature-Importance Analysis

Feature-importance analysis of XGBoost showed that the prediction was dominated by recent historical demand. The three most influential features were **load_lag_1**, **load_roll_mean_4**, and **load_lag_2**, with respective importance values of approximately **0.5066**, **0.2484**, and **0.2331**. Together, these features accounted for approximately **98.8% of the model's total feature importance**.

This indicates that recent load behavior is substantially more influential for short-term demand forecasting than individual renewable-generation variables. The result is consistent with the highly persistent nature of electricity demand.

B. Renewable-Reliability Results

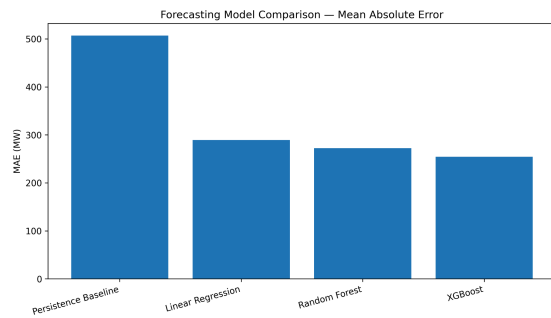
In addition to forecasting, the framework analyzed the relationship between electricity demand and renewable availability. Renewable generation was aggregated from solar, wind, and hydroelectric components, and renewable coverage was calculated relative to electricity demand. Observations were categorized into four demand–renewable conditions:

Demand	Renewability	Interpretation
Low	Low	Lower demand pressure
Low	High	Favorable renewable condition
High	Low	Potentially critical condition
High	High	High demand with renewable support

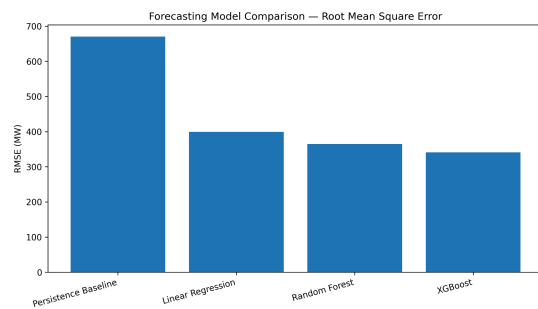
Table 2: Demand vs Renewability confusion matrix

Using the defined demand and renewable thresholds, the framework identified **194 potentially critical reliability windows** characterized by high electricity demand combined with comparatively low renewable availability.

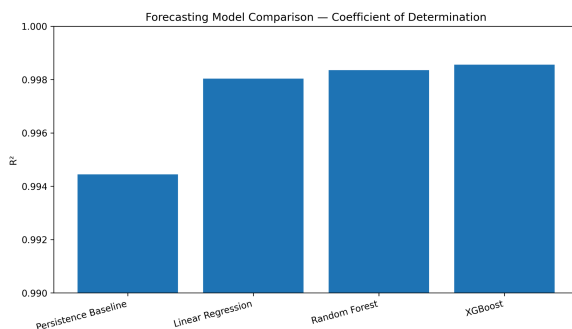
C. Visualizations



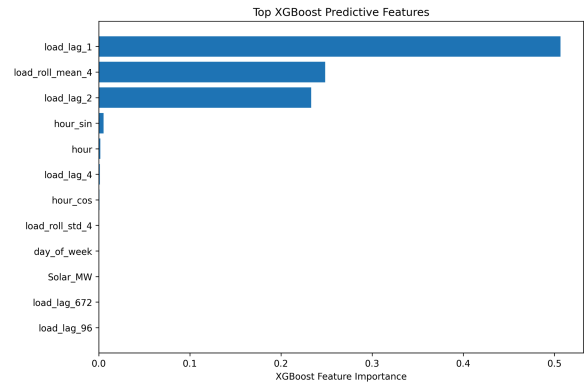
Graph 8.3.1: Model Comparison based on MAE



Graph 8.3.2: Model Comparison based on RMSE



Graph 8.3.3: Model Comparison based on R²



Graph 8.3.4: XGBoost Predictive Features.

IX. DISCUSSION

The experimental results demonstrate that machine-learning models can effectively forecast short-term electricity demand using historical load and temporal characteristics. XGBoost produced the lowest MAE and RMSE among all evaluated models, indicating that its nonlinear ensemble structure was able to capture relationships that were not completely represented by Linear Regression. Random Forest also performed strongly, while Linear Regression provided a useful baseline for evaluating the additional benefit obtained from nonlinear learning.

The persistence baseline achieved an R² of **0.994445**, confirming the strong temporal dependence of the German electricity-demand series. Therefore, the high R² obtained by the machine-learning models should not be interpreted as near-perfect prediction in an absolute sense. Instead, it reflects the fact that electricity demand at 15-minute intervals is highly dependent on recent observations. The more informative measures for comparing the models are therefore MAE and RMSE, where XGBoost demonstrated substantial improvement over persistence.

The approximately **49.8% reduction in MAE** achieved by XGBoost relative to persistence indicates that the model extracts useful information beyond simply carrying forward the previous observation. The dominance of **load_lag_1**, **load_roll_mean_4**, and **load_lag_2** in feature importance further indicates that recent load behavior is the primary source of predictive information. Temporal features such as hour and day of week provide additional contextual information, while the individual renewable-generation variables contribute comparatively less to the direct load-forecasting task.

An important distinction is observed between the **forecasting component** and the **renewable-reliability component**. Renewable-generation information is highly relevant for evaluating system adequacy and identifying periods in which high demand coincides with low renewable availability. However, target-derived quantities such as renewable coverage and renewable gap were excluded from the forecasting feature matrix to prevent target leakage. Consequently, the forecasting

results represent a cleaner evaluation of historical and temporal predictors rather than a model receiving information mathematically derived from the target.

The identification of **194 potentially critical windows** demonstrates the additional value of combining forecasting with renewable-reliability analysis. Instead of considering electricity demand independently, the proposed framework provides an analytical mechanism for identifying situations where high demand occurs simultaneously with comparatively low renewable support. Such periods can be useful for operational planning, reserve management, and future integration of renewable resources.

X. CONCLUSION

This work presented a data-driven framework for **short-term electricity-load forecasting and renewable-reliability analysis using German electricity data** covering three years at 15-minute temporal resolution. The proposed methodology incorporated systematic data validation, temporal and historical feature engineering, chronological model evaluation, renewable-generation analysis, and demand-renewable condition classification.

Among the evaluated forecasting approaches, **XGBoost achieved the best performance**, with an MAE of **254.36 MW**, RMSE of **341.28 MW**, and $R^2=0.998561$. It outperformed Random Forest, Linear Regression, and the persistence baseline in terms of prediction error. The persistence baseline itself achieved $R^2=0.994445$, demonstrating the strong temporal persistence inherent in electricity demand.

Feature-importance analysis showed that recent historical load observations were the dominant predictors, while the renewable analysis provided a complementary reliability perspective. The framework identified **194 potentially critical windows** corresponding to high-demand and low-renewable-availability conditions.

Overall, the study demonstrates that combining **short-term load forecasting with renewable-availability analysis** can provide a more informative representation of electricity-system behavior than forecasting demand alone. The methodology also establishes a reproducible foundation for extending the framework toward more advanced forecasting and energy-system decision-support applications.

XI. FUTURE WORK

Several extensions can further improve the proposed framework.

1. Longer-horizon forecasting:

The present formulation focuses on short-term forecasting using 15-minute observations. Future work can investigate **1-hour, 6-hour, day-ahead, and multi-day forecasting**, where the predictive importance of historical and renewable features may differ substantially.

2. Weather and exogenous variables:

Temperature, humidity, solar irradiance, wind speed, precipitation, holidays, and special events can be incorporated to capture external factors affecting electricity demand and renewable generation.

3. Advanced deep-learning models:

Future experiments can evaluate **LSTM, GRU, Temporal Convolutional Networks, Transformers, and Temporal Fusion Transformers** to determine whether advanced sequence models can provide improvements over tree-based approaches.

4. Probabilistic forecasting:

Instead of generating a single deterministic prediction, future versions can produce prediction intervals or probability distributions. This would allow system operators to quantify forecasting uncertainty and make risk-aware decisions.

5. Renewable-aware forecasting:

Future work can investigate forecast-time renewable availability more deeply by incorporating **genuine renewable forecasts or weather-derived estimates available before the prediction horizon**, rather than using contemporaneous observations.

6. Reliability and risk scoring:

The identified high-demand/low-renewable windows can be extended into a continuous reliability-risk score using demand magnitude, renewable deficit, duration, and forecast uncertainty.

7. Real-time deployment:

The complete pipeline can eventually be deployed as a real-time monitoring dashboard capable of updating load forecasts and automatically identifying potentially critical operating periods.

8. Comparative regional analysis:

The methodology can be extended beyond Germany to other European electricity markets to investigate whether the observed forecasting behavior and renewable-reliability patterns remain consistent across regions.

XII. REFERENCES

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